

Graduation Project Proposal

Sign Language Recognition Glove Using Sensor Data and Transfer Learning

1. Project Overview

This project proposes the design and implementation of a **smart hand glove** capable of recognizing **sign language letters** by reading finger positions and hand gestures, then converting them into **spoken words using a speaker**. The system aims to assist communication for people with hearing or speech impairments.

2. Problem Statement

Most sign language recognition systems rely on **camera-based vision models**, which suffer from limitations such as lighting conditions, camera angles, and privacy concerns. On the other hand, **sensor-based gloves** provide more stable and private data. However, a major challenge is the **lack of large, labeled datasets based on glove sensor inputs**, which makes training accurate machine learning models difficult.

3. Proposed Solution

To overcome the dataset limitation, this project adopts a **transfer learning approach**:

- Existing **hand gesture image datasets** will be used as the primary data source.
- Hand images will be processed using a **hand landmark detection model** (e.g., MediaPipe).
- Each gesture will be converted into **vector representations of finger joint positions relative to the palm**.
- These vectors will serve as a **synthetic sensor-like dataset**.
- A machine learning model will be trained on this data.
- The trained model will then be **adapted and fine-tuned using real sensor data** collected from the glove.

4. Scope Limitation

Sign language **words and sentences often involve arm, body, and facial movements**, which are outside the capability of a hand glove alone.

Therefore, the scope of this project is **limited to recognizing individual letters only**, as letters rely mainly on **finger positions and hand posture**, making them suitable for glove-based recognition.

5. System Components

- **Hardware**
 - Smart glove with finger-bend sensors (e.g., flex sensors)
 - Microcontroller (Arduino / ESP32)
 - Speaker module for audio output
- **Software**
 - Hand landmark extraction from image datasets
 - Feature vector generation (relative joint positions)
 - Machine learning model using transfer learning
 - Fine-tuning with real sensor data
 - Text-to-speech module

6. Methodology

1. Select an existing hand gesture image dataset.
2. Extract hand landmarks and convert them into relative vector features.
3. Train a classification model on the generated vectors.
4. Collect real-time sensor data from the glove.
5. Fine-tune the trained model to adapt to sensor input.
6. Deploy the model on embedded hardware.
7. Convert recognized letters into audio output via a speaker.

7. Expected Outcomes

- A functional **sign language glove** capable of recognizing letters.
- A **transfer learning pipeline** that bridges image-based data and sensor-based data.
- Improved accuracy compared to training solely on limited sensor datasets.
- A scalable foundation for future extension to words or sentences using additional sensors.

8. Significance of the Project

This project demonstrates an innovative use of **transfer learning** to solve real-world data scarcity problems. It combines **embedded systems, machine learning, and assistive technology**, making it both technically challenging and socially impactful.